

Question

In general atomic fullerene inclusion compounds, atoms can either form bonds with the fullerene or have no bonding interaction with the fullerene. Reacting K_3As_7 , $\text{Ni}(\text{COD})_2$, Bu_4PBr in ethylenediamine solution yields a precipitate of tetrabutylphosphonium salt. Single-crystal diffraction confirms that the structure of the anion in this salt can be represented as $[\text{As}@\text{Ni}_m@\text{As}_n]^{3-}$. (The structure before the @ symbol is "included" within the structure after the symbol.) This anion has the same symmetry as a regular icosahedron and consists solely of As and Ni elements, with a nickel mass fraction of 30.92%. Treating the outermost As_n layer as an independent molecule,

Given: the covalent radius of arsenic is 119 pm, and its van der Waals radius is 185 pm; the nickel atoms in the Ni_m cluster are in contact with each other, and the distance between the farthest-spaced nickel atoms is 516 pm, while the van der Waals radius of Ni is 163 pm; the As-As distance in As_n is 275.2 pm, and the distance between the farthest-spaced arsenic atoms is 771.2 pm.

Calculate the distance between each vertex Ni atom in the Ni_m cluster and the nearest As atom on the As_n layer, and select the closest answer.

A. 239 pm

B. 234 pm

C. 306 pm

D. 307 pm

E. 613 pm

F. 163 pm

G. 185 pm

H. 239 pm

I. 246 pm

Answer

Correct Answer: **H**

Detailed Explanation

First, it is necessary to deduce the structure of this fullerene inclusion compound, i.e., to determine the values of m and n .

Since the mass fraction of nickel in the anion is known, the ratio of the number of arsenic atoms to nickel atoms in the anion can be derived as $(0.3092/58.69) : ((1 - 0.3092)/74.92) = 1 : 1.75 = 4 : 7$.

CHECKPOINT

1 PTS

The ratio of arsenic atoms to nickel atoms in the anion is 4:7

The anion has the same symmetry as a regular icosahedron, so the m vertices and n vertices should form a polyhedron consistent with icosahedral symmetry. In practice, only the regular dodecahedron and icosahedron satisfy this condition.

CHECKPOINT

1 PTS

The m vertices and n vertices should form a polyhedron consistent with icosahedral symmetry

Thus, there are at least 12 nickel atoms, with 21 arsenic atoms, where 20 arsenic atoms form a regular icosahedron. Therefore, m is 12 and n is 20.

CHECKPOINT

1 PTS

m is 12, n is 20

To satisfy the symmetry of a regular icosahedron, the vertices of the Ni polyhedron must align with the face centers of the As polyhedron.

CHECKPOINT

1 PTS

The vertices of the Ni polyhedron must align with the face centers of the As polyhedron.

Let the distance between the farthest face centers of the As_{20} polyhedron be d , and the distance from a vertex to the face center of each As_5 face be r .

Then

$$r = (275.2/2)/\sin 36^\circ \text{pm} = 234.1 \text{pm}$$

CHECKPOINT

1 PTS

The distance from a vertex to the face center of each As_5 face is 234.1 pm

$$d/2 = \left[(771.2/2)^2 - 234.1^2 \right]^{1/2} \text{pm} = 306.4 \text{pm}$$

$$d = 612.8 \text{pm}$$

The distance from a nickel atom to the nearest face center of the As_{20} polyhedron is

$$(612.8 - 516)/2\text{pm} = 48.4\text{pm}$$

CHECKPOINT

1 PTS

The distance from a nickel atom to the nearest face center of the As_{20} polyhedron is 48.4 pm

Based on the right triangle relationship, $d(\text{Ni} - \text{As}) = (48.4^2 + 234.1^2)\text{pm} = 239\text{pm}$

CHECKPOINT

1 PTS

$$d(\text{Ni} - \text{As}) = (48.4^2 + 234.1^2)\text{pm} = 239\text{pm}$$

Thus, option H is correct.